

Deployment

Part 2: Setting up AWS Infrastructure

In the previous videos...

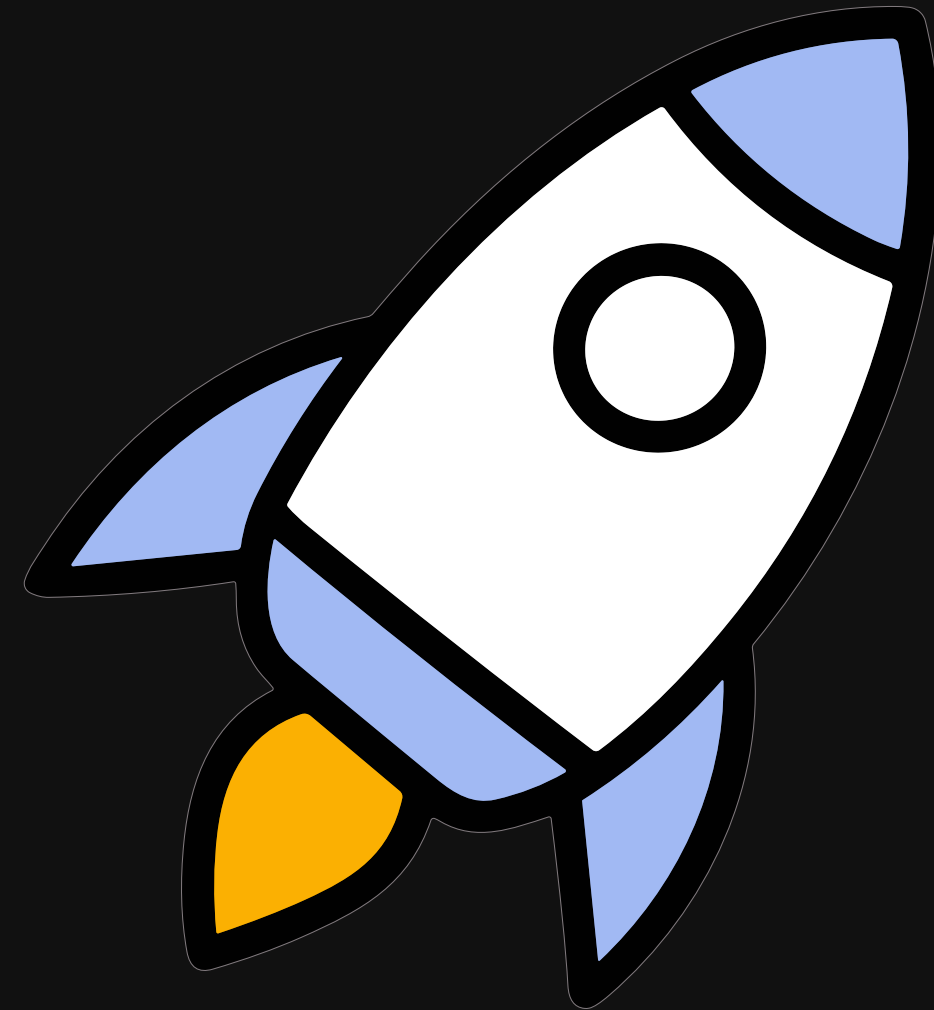
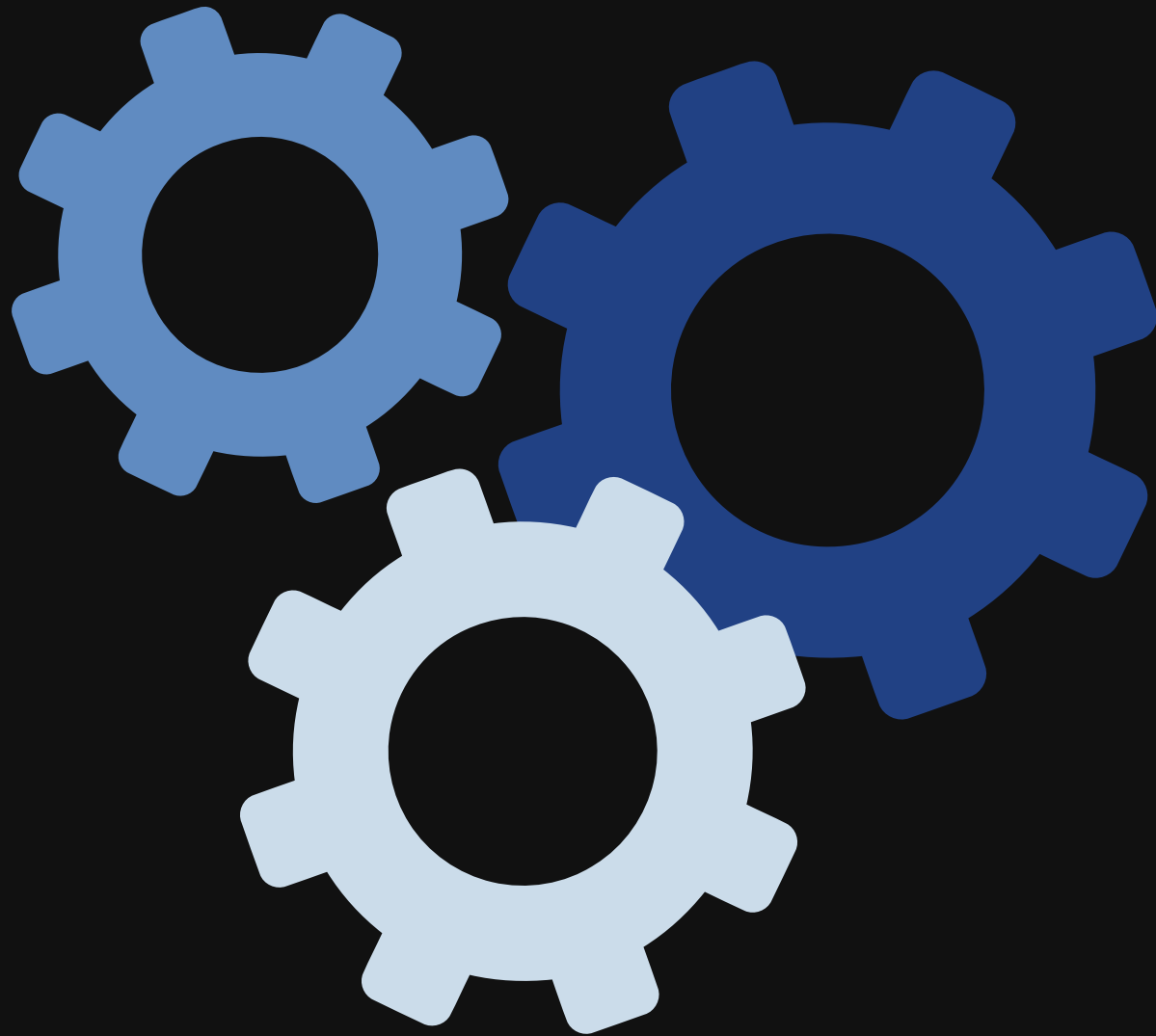


In this video...

- Launch an EC2 instance
- Configure the Security Group



Launch an EC2 instance





Console Home Info

Reset to default layout

+ Add widgets

Recently visited Info



No recently visited services

Explore one of these commonly visited AWS services.

[EC2](#) [S3](#) [CloudWatch](#) [IAM](#)

[View all services](#)

Applications (0) Info

Create application

Region: Europe (Stockholm)

Select Region

eu-north-1 (Current Region) ▾

Q Find applications

< 1 >

Name ▾	Description ▾	Region ▾	Originati. ▾
--------	---------------	----------	--------------

✖ ▶ Access denied to servicecatalog:ListApplications

 Diagnose with Amazon Q

[Go to myApplications](#)

Welcome to AWS



Getting started with
AWS ↗
Learn the fundamentals and

AWS Health Info

Cost and usage Info

Current month
✖ Access denied

Cost breakdown
✖ Access denied

- Services
- Features
- Documentation
- Knowledge articles
- Marketplace
- Blog posts
- Events
- Tutorials

Services

Show more

- 

EC2

Virtual Servers in the Cloud


- 

EC2 Image Builder

A managed service to automate build, customize and deploy OS images


- 

Recycle Bin

Protect resources from accidental deletion



Features

Show more

- EC2 Instances

CloudWatch feature
- EC2 Resource Health

CloudWatch feature
- Dashboard

EC2 feature

Were these results helpful?

 Yes

 No

Documentation

Show more



EC2



Dashboard

[AWS Global View](#)

[Events](#)

Instances

[Instances](#)

[Instance Types](#)

[Launch Templates](#)

[Spot Requests](#)

[Savings Plans](#)

[Reserved Instances](#)

[Dedicated Hosts](#)

[Capacity Reservations](#)

[Capacity Manager](#) New

Images

[AMIs](#)

[AMI Catalog](#)

Elastic Block Store

[Volumes](#)

[Snapshots](#)

[Lifecycle Manager](#)

⚠ Unable to read or write preferences from local storage. Enable local storage in your browser and opt into the functional cookies preference to use the default landing page feature. This can be changed in the bottom right of the page footer.



Resources



You are using the following Amazon EC2 resources in the Europe (Stockholm) Region:

Instances (running)	0	Auto Scaling Groups	0	Capacity Reservations	0
Dedicated Hosts	0	Elastic IPs	0	Instances	0
Key pairs	0	Load balancers	0	Placement groups	0
Security groups	1	Snapshots	0	Volumes	0

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

Launch instance



[Migrate a server](#)

Note: Your instances will launch in the Europe (Stockholm) Region

Service health

[AWS Health Dashboard](#)



✖ An error occurred
An error occurred retrieving service health information

[Diagnose with Amazon Q](#)

Zones

EC2 cost



Date range: Past 6 months

Region: Global

Total cost

Unable to load

Regions

Unable to load

Unable to load

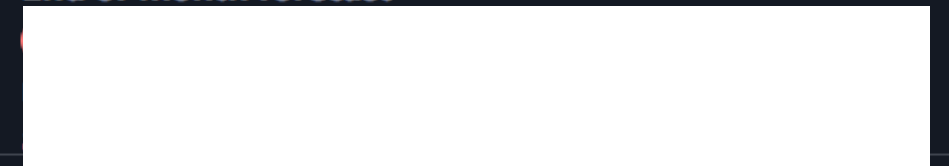
[Analyze your costs in Cost Explorer](#)

EC2 Free Tier [Info](#)

Offers for all AWS Regions.

0 EC2 free tier offers in use

End of month forecast



It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Take a walkthrough

[Do not show me this message again.](#)

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

calculator-server

[Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Quick Start

Amazon
Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

aws



[Browse more AMIs](#)

Including AMIs from
AWS Marketplace and

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd64...[read more](#)
ami-091138d0f0d41ff90

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

[Preview code](#)

▼

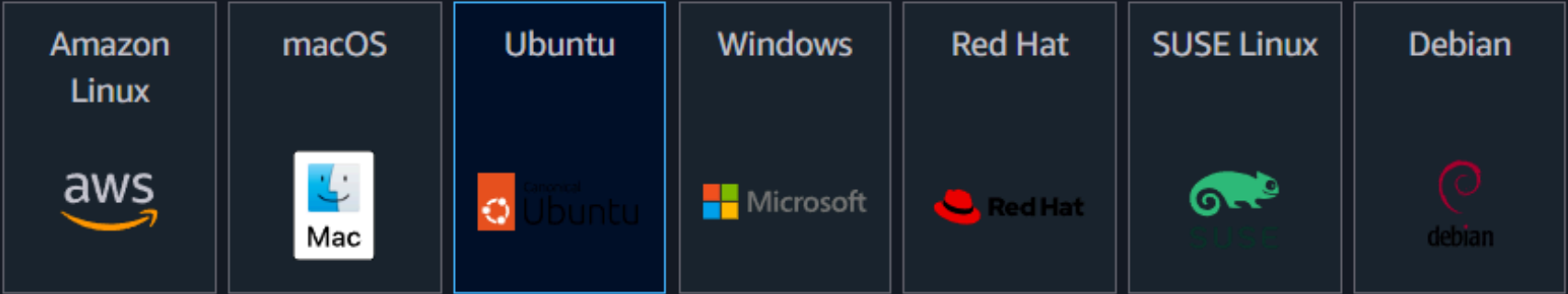
Application and OS Images (Amazon Machine Image)

Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Quick Start



Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 26.04 LTS (HVM), SSD Volume Type

Free tier eligible

ami-091138d0f0d41ff90 (64-bit (x86)) / ami-07ad186bc37b8dac4 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 26.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 26.04, amd64 resolute image

Architecture	AMI ID	Publish Date	Username	
64-bit (x86)	ami-091138d0f0d41ff90	2026-04-21	ubuntu	Verified provider

▼ Summary

Number of instances

Info

1

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd64...[read more](#)

ami-091138d0f0d41ff90

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure

Key pair name - *required*

Select

▼ Network settings [Info](#)

Network [Info](#)

vpc-0e5afb5130e62b60c

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance.

☒ Create security group

☐ Select existing

We'll create a new security group called 'launch-wizard-1' with the

☒ Allow SSH traffic from

Anywhere

Create key pair



Key pair name

Key pairs allow you to connect to your instance securely.

calculator-key-pair

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#) ↗

Cancel

Create key pair

Summary

Number of instances [Info](#)

Software Image (AMI)

Amazon Linux 2, Ubuntu, 26.04, amd64...[read more](#)
ami-091138d0f0d41ff90

Instance type (instance type)

t3.micro

Firewall (security group)

Default security group

Storage (volumes)

Volume(s) - 8 GiB

Cancel

Launch instance

🗨 Preview code

▼ Configure storage [Info](#)

Advanced

1x GiB ▼ Root volume, 3000 IOPS, Not encrypted

Add new volume

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

🕒 Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.



File systems

☐ S3 Files - new

☐ EFS

☐ FSx

☒ None

▶ Advanced details [Info](#)

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd64...[read more](#)
ami-091138d0f0d41ff90

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

🔍 Preview code

✔ **Success**
Successfully initiated launch of instance (i-XXXXXXXXXX)

▶ Launch log

Next Steps

🔍 What would you like to do next with this instance, for example "create alarm" or "create backup"

< 1 2 3 4 5 6 >

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

[Create billing alerts](#) ↗

Connect to your instance

Once your instance is running, log into it from your local computer.

[Connect to instance](#) ↗

[Learn more](#) ↗

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

[Connect an RDS database](#) ↗

[Create a new RDS database](#) ↗

[Learn more](#) ↗

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

[Create EBS snapshot policy](#) ↗

[Manage detailed monitoring](#)

[Create Load Balancer](#)

[Create AWS budget](#)

[Manage CloudWatch alarms](#)

EC2 <

- Dashboard
- AWS Global View ↗
- Events

▼ Instances

Instances

- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations
- Capacity Manager [New](#)

▼ Images

- AMIs
- AMI Catalog

▼ Elastic Block Store

- Volumes
- Snapshots

Instances (1) Info

[Connect](#) [Instance state ▼](#) [Actions ▼](#) [Launch instances ▼](#)

Saved filter sets
Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

< 1 > ⚙

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	F
☐	calculator-server	i-██████████	✓ Running	t3.micro	⌚ Initializing	us-east-1c	e-██████████om...	5

Select an instance

⚙ ▼

[EC2](#)[Instances](#)

i-



EC2

[Dashboard](#)[AWS Global View](#)[Events](#)

Instances

[Instances](#)[Instance Types](#)[Launch Templates](#)[Spot Requests](#)[Savings Plans](#)[Reserved Instances](#)[Dedicated Hosts](#)[Capacity Reservations](#)[Capacity Manager](#)[New](#)

Images

[AMIs](#)[AMI Catalog](#)

Elastic Block Store

[Volumes](#)[Snapshots](#)[Lifecycle Manager](#)

Instance summary for i- (calculator-server) [Info](#)

Updated less than a minute ago

Instance ID



i-

IPv6 address

-

Hostname type

IP name: ip-

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address



[Public IP]

IAM role

-

IMDSv2

Required

Public IPv4 address

[open address](#)

Instance State

Running

Private IP DNS name (IPv4 only)



ip-

Instance type

t3.micro

VPC ID



vpc-

Subnet ID



subnet-

Instance ARN

[Connect](#)[Instance state](#)[Actions](#)

Private IPv4 addresses



Public DNS

[open address](#)

Elastic IP addresses

-

AWS Compute Optimizer finding

[Retry](#)

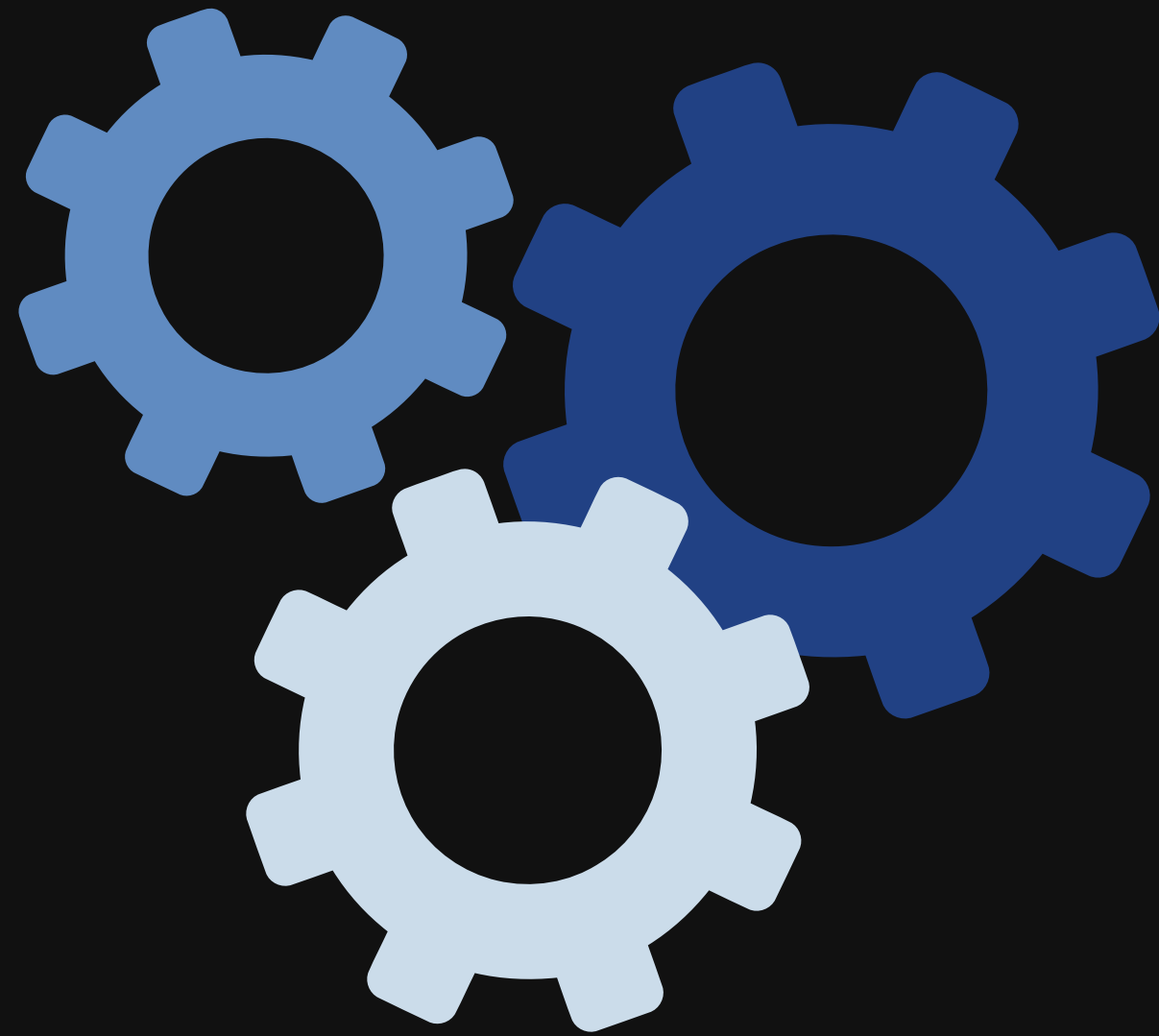
Auto Scaling Group name

-

Managed

false

Configure the Security Group



- EC2
- Dashboard
- AWS Global View ↗
- Events
- ▼ Instances
 - Instances
 - Instance Types
 - Launch Templates
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 - Capacity Manager [New](#)
- ▼ Images
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- ▼ Elastic Block Store
 - Volumes
 - Snapshots

Details | Status and alarms | Monitoring | **Security** | Networking | Storage | Tags

▼ Security details

IAM role
-

Owner ID
 [redacted]

Launch time
Sun May 31 2026 11:09:53 GMT+0200 (South Africa Standard Time)

Security groups

 sg-[redacted] (launch-wizard-1)

▼ Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups
-	sg-[redacted]	22	TCP	0.0.0.0/0	launch-wizard-1

▼ Outbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Destination	Security groups
-	sg-[redacted]	22	TCP	0.0.0.0/0	launch-wizard-1

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 - Volumes
 - Snapshots

sg-[redacted] - launch-wizard-1

Actions

Details

Security group name launch-wizard-1	Security group ID sg-[redacted]	Description launch-wizard-1 created 2026-05-31T09:02:48.495Z	VPC ID vp-[redacted]
Owner [redacted]	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

- Inbound rules
- Outbound rules
- Sharing
- VPC associations
- Related resources - new
- Tags

Inbound rules (1)

Manage tags Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-[redacted]	IPv4	SSH	TCP	22

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID

Type Info

Protocol Info

Port range

Source Info

Description - optional Info

sgr-0a319d9339815b417

SSH

TCP

22

Custom

Q

0.0.0.0/0 X

Delete

-

HTTP

TCP

80

Anywh...

Q

0.0.0.0/0 X

Delete

-

Custom TCP

TCP

5000

Anywh...

Q

0.0.0.0/0 X

Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

X

Cancel

Preview changes

Save rules

Next Step

SSH in and Install Docker